Introduction

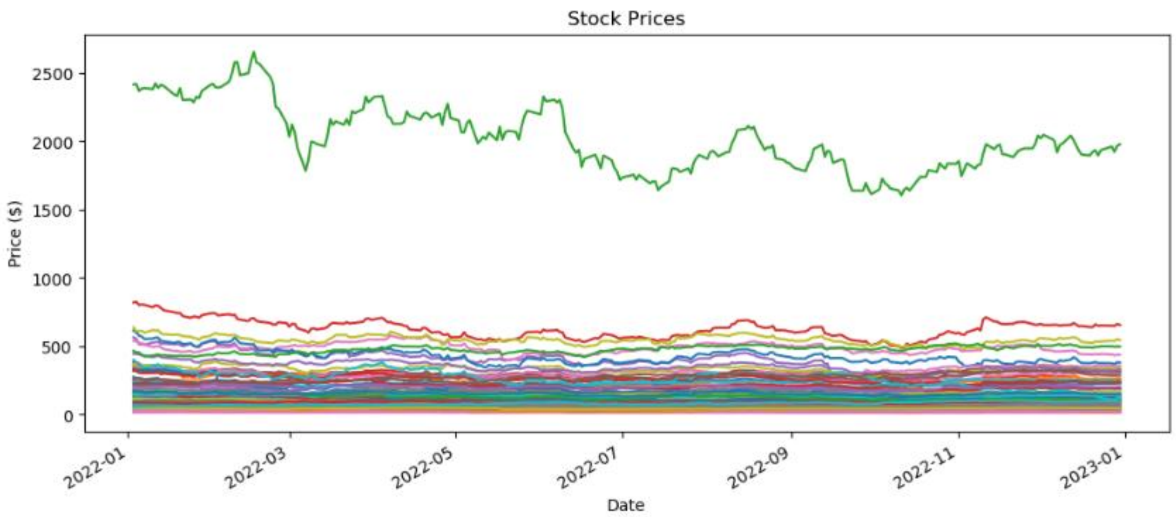
This report analyses the behavior of stock market prices for 70 top performing companies publicly traded over a 10-year period from 2015-2025; data was obtained using Yahoo Finance API. The report explores key characteristics of financial markets including returns, volatility, correlations, drawdown, and return distributions. Statistical analysis and visualization are used to examine how different companies differ in terms of risk and performance. Additionally, an unsupervised machine learning algorithm (K-Means clustering) is performed and used to group companies with similar financial characteristics, giving insight into how stocks behave within the broader market

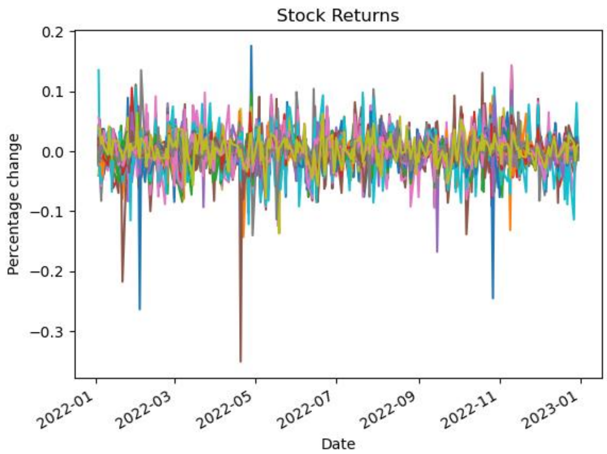
Data_ananual_return_volatility (1)

Using the yahoo finance library, 70 companies' stock data was analyzed. Data was downloaded from the beginning of 2015 to the end of 2024 (10-year period). Metadata includes; Close Price, Daily High, Daily Low, Open Price, Volume Traded, and Date as the shared index. Any companies that had less than 90% of days' worth of data over the last 10 years were dropped from the datasheet.

Stock Prices and Returns

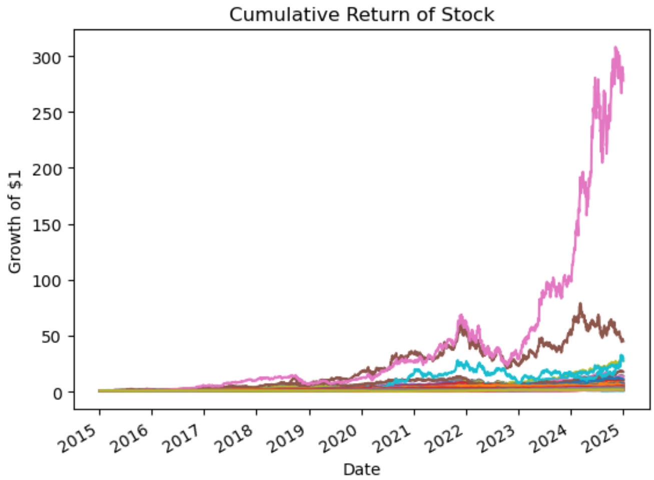
All 70 companies were plotted to analyze trends in daily stock prices (using close prices) and returns. Data here is messy and no legend is included since it wouldn't fit in graph, so here on out only 1 companies data set will be analyzed and plotted at a time.





Cumulative Returns

Cumulative returns of each stock were calculated using the daily return rate (over 10 years), and top 10 highest returns were found.



Company	Cumulative Return
Nvidia	277.95
AMD	45.24
Broadcom	30.39
Arista Network	28.21
Tesla	27.62
Netflix	17.88
Amazon	14.22
Eli Lilly	13.33
Progressive	11.37
Microsoft	10.50

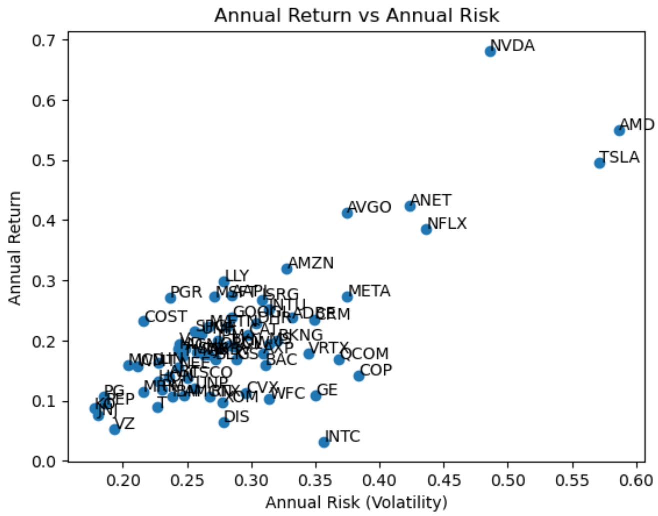
****Note**: Data here is messy and no legend is included since it wouldn't fit in graph, so here on out only 1 companies data set will be analyzed and plotted at a time.

Annual Return and Volatility

Using the mean returns and mean volatility over the 10-year period, then formatting these values as a DataFrame.

Risk, Correlations, and Drawdown (2)

Using the DataFrame constructed containing Averaged Annual return and volatility, companies were plotted on a scatter plot. Visualizing Return Vs Volatility, it was observed that most companies cluster in the region of low risk - low return with the outlier in the high risk - high return region.



Note: Companies of low risk – high return are ideal however no companies were in this region.

<u>Correlation Heatmap</u>

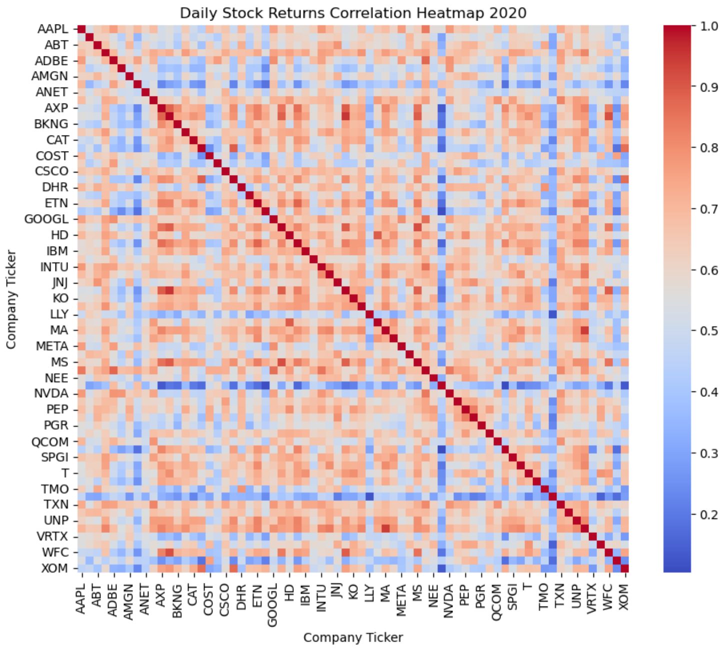
Correlation heat maps show how companies move together, specifically their daily stock returns. Heatmaps use color along with a color legend to quantify the relation:

> +1 Moves perfectly

> 0 No relationship

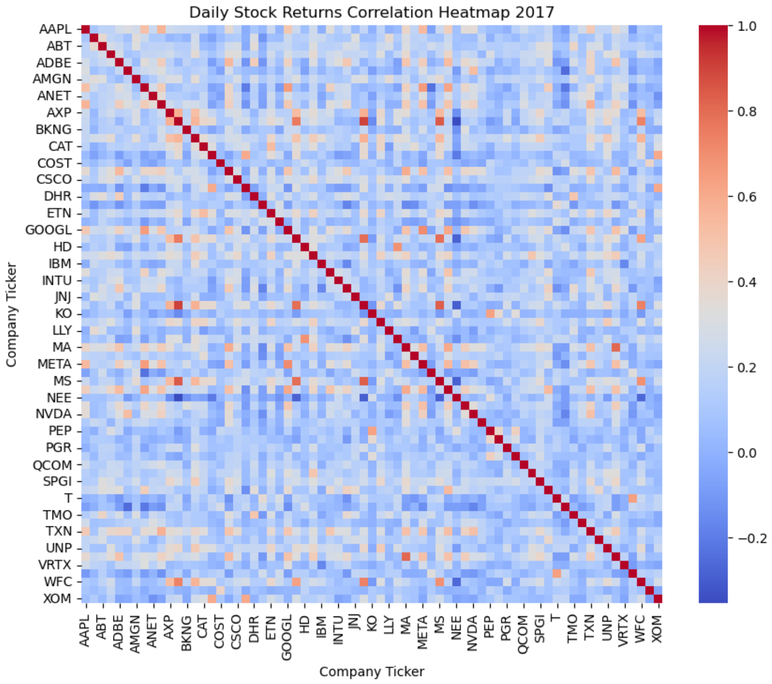
> -1 Moves Perfectly Opposite

Firstly, a correlation heatmap was generated using 2020 daily stock returns, this was to highlight the effect of a crises on the financial markets:



When the COVID pandemic began in March 2020, there was heavy stress on the market and the general trend of all companies (as indicated by the S&P500) was a negative return rate. During this time (crises), all stocks tend to have high correlation indicated by more red in the diagram below. This shows that although companies may not be in the same industry or share a marketplace/customers, during crises/stress they are highly correlated.

Note: It is interesting to note some stocks remain highly anti-correlated to the market, notable companies include (Netflix and Tesla). This indicates that certain companies can still perform well during market stress times.



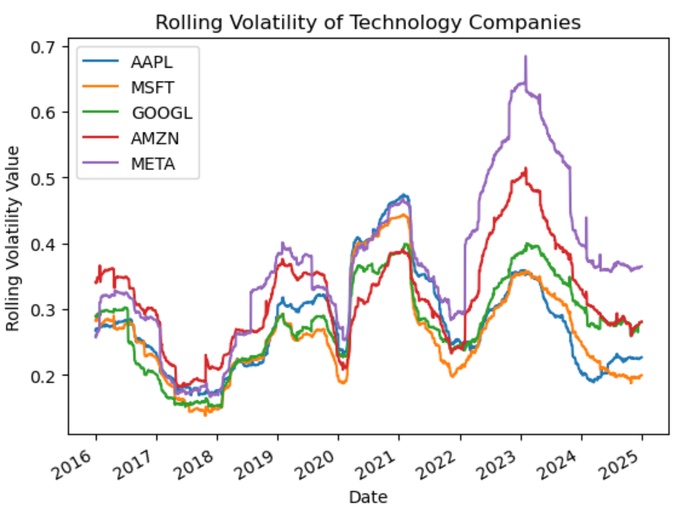
Then using data from a stable market (2017), a correlation heatmap was created. It is clear to see that there is much less correlation during these times, and most companies are anti-correlated (indicated by blue).

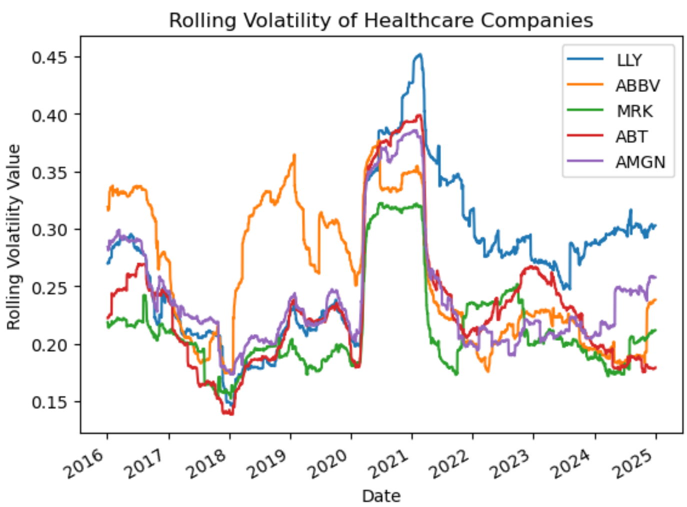
Note: The same companies that were anti-correlated to most of the market in 2020 crisis are not also anti-correlated to the market during 2017 times. Indicating that these companies are not permanently anti-correlated, there may be factors during crises times which lead to these companies 'doing well'.

<u>Rolling Volatility</u>

A 252 day rolling volatility was calculated for 5 Technology and 5 Healthcare companies to analyse volatility trends that industries follow over the 10 year period.

The aim of this is to show that companies within the same sector show similar volatility trends, and act differently to the market as a whole.





Technology companies undergo stress at similar times this is shown as high volatility where all companies are correlated with one another and have similar trends.

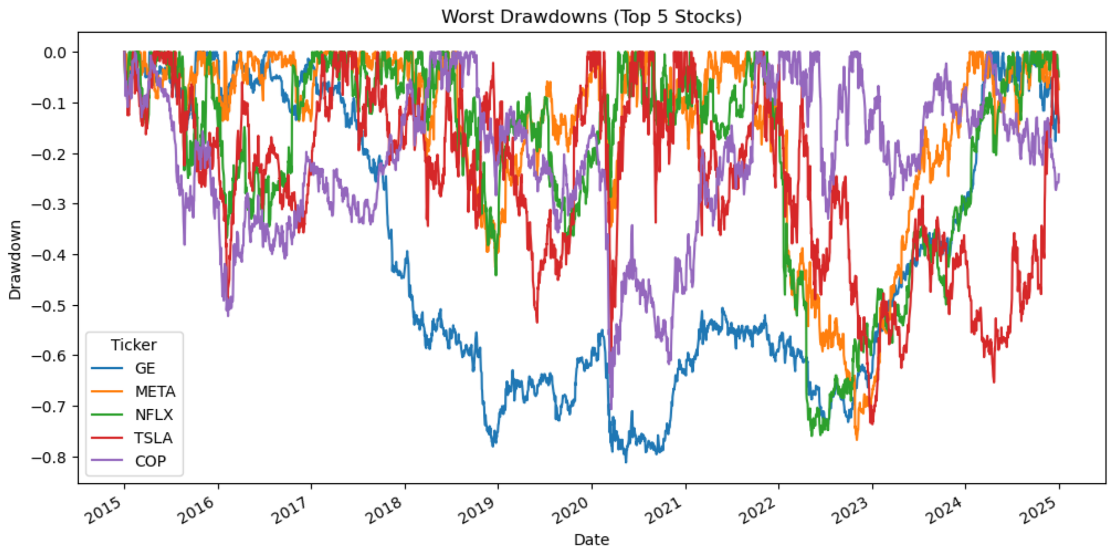
The same can be said for healthcare companies, however some remain more volatile than other for longer periods of time.

Important Note: Although within each of the graphs, companies within the same industry show correlated volatility, this however does not mean the two graphs themselves follow similar trends. i.e. Volatility in one sector does not correlate to volatility in another sector.

Drawdowns

Drawdowns are key in identifying "Psychological risk", while returns show absolute values drawdowns tell us how far a stock has fallen from its highest values previously (the highest value is updated as time progresses and higher peaks are reached). Drawdowns are calculated as a percentage deviation from the previous highest value of the stock.

All maximum drawdowns over the last 10 years (for all companies) were evaluated and the top 5 'worst' (largest in magnitude) drawdowns were identified and plotted.



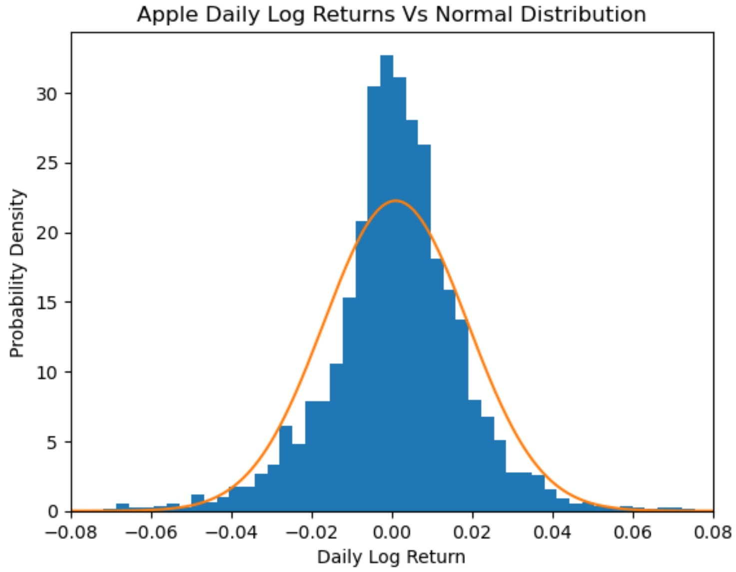
Top 5 Worst Drawdown from the last 10 years

Company	Date	Drawdown (%)
General Electric (GE)	2020-05-15	-0.81
Meta	2022-11-03	-0.77
Netflix	2022-05-11	-0.76
Tesla	2023-01-03	-0.74
ConocoPhillips (COP)	2020-03-18	-0.71

Returns Distributions (3)

Log returns are used as standard in finance, here a normal distribution is compared with the empirical data (as log returns) to compare and test whether returns follow a normal (log-normal) distribution.

Using the log returns' mean and standard deviation of Apple's stocks we overlay a normal gaussian distribution over a histogram of the log returns distribution.



Comparisons:

1. The empirical centre is more peaked than the normal distribution
2. Tails of the distribution are heavier than the normal distribution

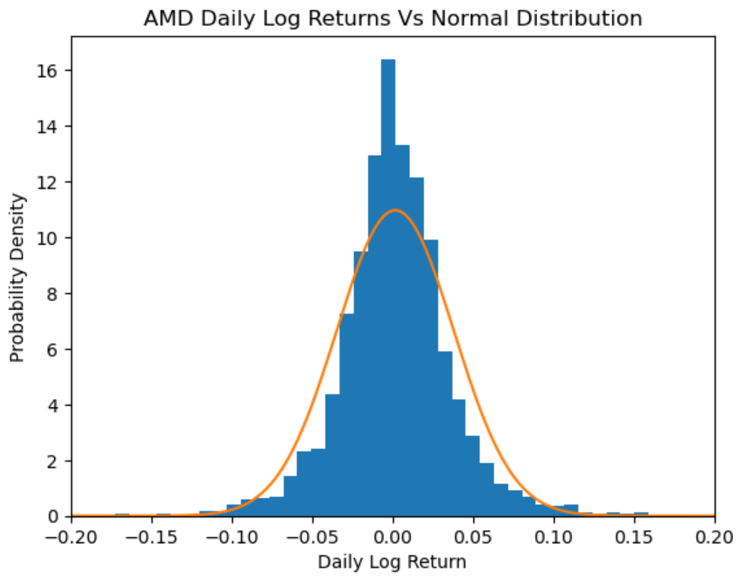
A heavier tail suggests that events of 3σ (standard deviations), that should occur 0.3% of the time, in fact occur more frequently than a gaussian suggests (i.e. Crashes and spikes happens more often).

Kurtosis values are uses to measure the "tailedness" of a distribution, where a kurtosis value of 3 represents a gaussian and anything more is excess kurtosis signifying a fat tailed distribution.

The daily log returns of Apple stock showed a kurtosis value of ≈ 8, indicating fat tails and a sharp peak.

From the Risk Vs Returns scatter plot, AMD was identified to have large risk (volatility) so its distribution was explored.

AMD stock showed a similar high peak and fat tailed distribution like Apple's stocks, however it had a much larger kurtosis of 13.27, make returns further away from the mean more likely (this is indicative of high volatility/risk).

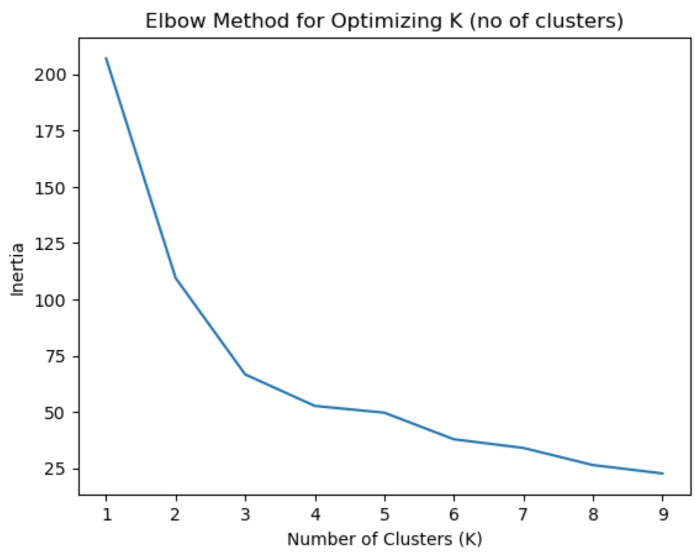


Machine Learning (4)

A K-Means unsupervised machine learning algorithm was used to determine groups (Clusters) of companies with similar trends. For this algorithm “features” (in the form of numbers) are required, for this various useful data was selected, Mean return, Volatility, and Max Drawdown. (**Note:** Next stage of development can use skewness as a feature).

For a chosen number of clusters, K-Means algorithm tries to minimize inertia (squared distance between scatter points and their cluster centre).

We can determine a suitable number of clusters for our features by trailing over numbers of cluster (1-10). Using the elbow method the desired number of clusters is found at the point where adding more clusters gives diminishing returns (minimizing inertia).



In this case there is a much smaller drop in inertia between 3-4 than there is 2-3, so we can use no of cluster = 3. A next stage in this would be to implement a silhouette fitting model, this again optimizes the number of clusters.

Cluster groups were then identified and similar features are explained:

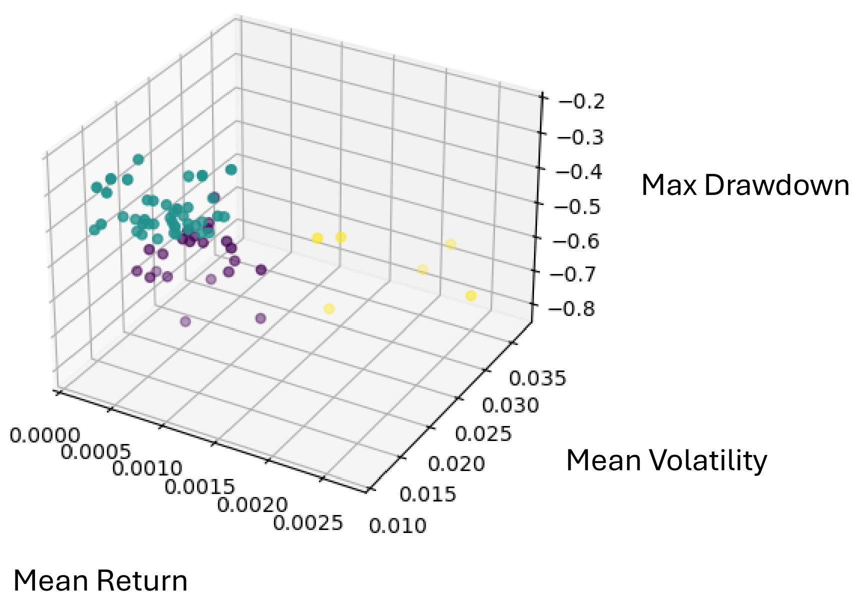
Cluster 0 - High returns, medium volatility, large drawdowns

Cluster 1 - High returns, low volatility, low volatility, low drawdowns

Cluster 2 - Low returns, High volatility, large drawdowns

These features were determined by taking a mean of returns, volatility, and max drawdowns over the clusters.

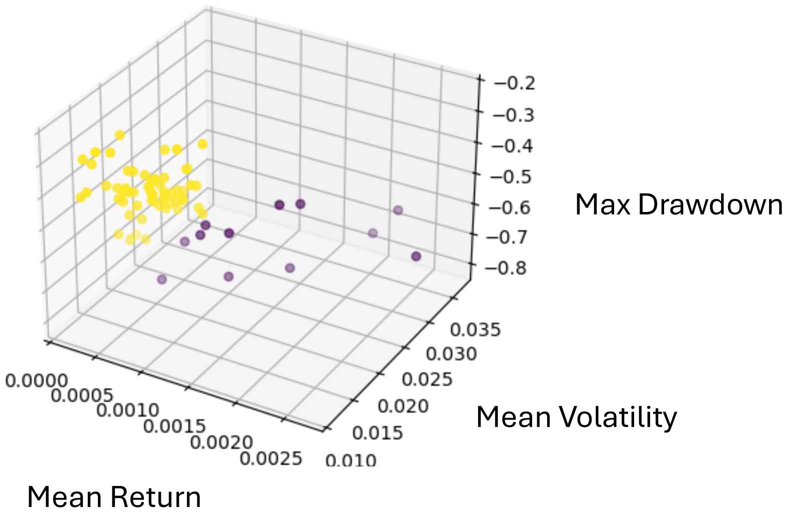
Clusters were then plotted at a scatter plot in the 3 dimensional features space (the same space that K-Means optimizes over).



Cluster no.	Mean Return	Mean Volatility	Max Drawdown
0	0.000683	0.020254	-0.57
1	0.000681	0.015446	-0.37
2	0.001950	0.030214	-0.64

The 3D plot shows clusters grouped by colour, visualising how clusters occupy similar spaces.

Results using 2 cluster groups for the K-Means Algorithm is also shown below (a silhouette model would help determine whether to use 2 or 3 clusters).



The data set is now broken into 2 cluster groups:

Cluster 1 - High Return, Medium volatility, Large Dragdown

Cluster 2 - Low Return, Low Volatility, Medium Dragdown

Cluster no.	**Mean Return**	**Mean Volatility**	**Max Drawdowns**
0	0.001411	0.026224	-0.65
1	0.000662	0.016587	-0.42

Conclusion

Overall, the analysis highlights several important properties of financial markets:

High return stocks tend to exhibit higher volatility and larger drawdowns, illustrating a fundamental trade-off between risk and return.

Correlation analysis shows that during periods of market stress, such as the COVID-19 crisis, stocks become more highly correlated, while in stable periods correlations weaken.

Return distribution analysis demonstrates that stock returns are not normally distributed and instead display fat tails, meaning extreme events occur more frequently than predicted by a Gaussian model.

Finally, K-Means clustering successfully grouped companies into distinct risk return profiles, showing that stocks with similar statistical characteristics tend to occupy similar regions in the features space.

These results demonstrate how statistical analysis and machine learning can be used to better understand patterns and structures within financial markets.